

Sriram Rajagopalan

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EDUCATION

London School of Economics and Political Science

MSc Economics

London, UK

Aug 2026 – Jun 2028

Madras School of Economics

B.A Economics (Honours)

Chennai, India

Aug 2023 – May 2026

- CGPA - 9.21/10.
- Relevant Modules - Calculus 1 and 2, Mathematical Statistics, Macroeconomics, Microeconomics, Time Series, Econometrics, Differential Equations, Stochastic Processes, and Real Analysis.

TEACHING EXPERIENCE

Madras School of Economics

Teaching Assistant

Chennai, India

Calculus 1

July 2024 – Nov 2024

- Facilitated weekly problem-solving sessions on functions, limits, continuity, differentiability, integration, and the Fundamental Theorem of Calculus, clarifying student doubts.

Mathematical Statistics

July 2025 – Nov 2025

- Conducted and graded weekly tests on probability distributions, point and interval estimation, and hypothesis testing; guided students through problem sets and doubts.

Intermediate Macroeconomics

Jan 2026 – May 2026

- Lectured on theories such as rational and adaptive expectations; solved weekly problem sets ranging from differential equations to endogenous growth theory.

QUANTITATIVE RESEARCH PROJECTS

Hamilton's Regime Switching Models

2025

- Authored a rigorous exposition of Markov-switching time-series models, deriving results from Neftçi (1984, *JPE*) and Goldfeld-Quandt (1973, *J. Econometrics*) through to Hamilton (1989).
- Formulated the finite-state Markov structure, the AR(1) representation of the latent regime, and the nonlinear filter and smoother.
- Coded the model in Python and recovered smoothed regime probabilities for US Real GNP, with recession dates closely tracking the NBER chronology.

Empirical Validity of the Phillips Curve in India

2026

- Tested the inflation-unemployment relationship using national annual time-series and an unbalanced state-level panel (rural and urban, 2017–23).
- Conducted Breusch–Pagan LM and F-tests (both indicating no significant panel effects); estimated pooled OLS with year dummies and state-clustered standard errors, plus an expectations-augmented (lagged-inflation) specification.
- Determined that the unemployment coefficient remained positive and insignificant - **no evidence of a Phillips curve**; inflation instead driven by supply-side (year) shocks and significant inflation persistence (lagged-inflation coefficient 0.27, $p = 0.01$).

Modelling US Recession Arrivals as a Poisson Process

2024 – 2026

- Co-authored an article in the college magazine testing a homogeneous Poisson model of post-war US recessions using quarterly FRED indicators.
- Evaluated fit via a Chi-Square goodness-of-fit test, a two-sample runs test, and ACF diagnostics.
- Programmed percentile, pivot, bootstrap- t , and parametric bootstrap confidence intervals in Python for the mean, variance, and variance-to-mean ratio.
- Detected sensitivity of arrival counts to the counting convention, motivating a renewal-process extension.

SKILLS

Programming Languages: Python (NumPy, Pandas, Matplotlib, statsmodels), Stata, and LaTeX.

Software: Microsoft Excel.

Other Academic Interests: Complex Analysis and Political Philosophy.